STAT210 Statistical Methods I: Linear Model

Lec 09: Multiple Linear Regression - SS

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Section 1

Review

Review: Lecture 01-06

- Lectures 01-06: SLR: least squares estimate, inference (CI, prediction, t-test, F-test), diagnostics, transformations, ANOVA, lack of fit test.
- Lecture 07: Multiple Linear Regression (MLR): least squares estimate.
- Lecture 08: MLR inference (CI, t-test, linear functions, ANOVA).
- Lecture 09: Sums of squares, more on F-tests and ANOVA.

Section 2

Overall F-test

F-test for the Overall Significance of MLR

- This test examines whether at least one predictor variable has a non-zero regression coefficient in the MLR model.
- The null and alternative hypotheses are:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$$
$$H_a : \text{at least one } \beta_j \neq 0, j = 1, 2, \ldots, p$$

- The F-statistic compares the fit of the full model (with all predictor variables) to the reduced model (with only the intercept).

The Full Model

- The ***full*** model is:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$$

- We denote the sum of squared errors (SSE) for the full model as SSE:

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

The Reduced Model

- The null hypothesis for the overall F-test in MLR is:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

- The ***reduced*** model under the null hypothesis is:

$$y_i = \beta_0 + \epsilon_i$$

- The sum of squares errors for the reduced model is denoted as $\sum_{i=1}^{n}(y_i - \bar{y})^2$, which is the SST.

The Difference in Fit between Full and Reduced Models

- We will decompose the total variability in the response variable y into two components

$$\begin{aligned} SST &= \sum_{i=1}^{n}(y_i - \bar{y})^2 \\ &= \underbrace{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}_{SSE} + \underbrace{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}_{SSR} \end{aligned}$$

- Here,
  - SST is the total sum of squares, calculated from the reduced model;
  - SSE is the error sum of squares, calculated from full model;
  - SSR is the regression sum of squares. Note $SSR = SST - SSE$.

The F-statistic for MLR

- The F-statistic tests the following hypotheses:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$$
$$H_a : \text{at least one } \beta_j \neq 0, j = 1, 2, \ldots, p$$

- The F-statistic is given by:

$$F = \frac{MSR}{MSE} = \frac{SSR/p}{SSE/(n-(p+1))}$$

where

- MSR is the mean square for regression,
- MSE is the mean square for error,
- p is the number of predictor variables,
- n is the number of observations.

The Distribution of the F-statistic

- Under H_0, $SSR/\sigma^2 \sim \chi^2_p$.
- $SSE/\sigma^2 \sim \chi_{n-p-1}$.
- SS and SSE are independent.
- Therefore, $F = \frac{MSR}{MSE} \overset{H_0}{\sim} F_{p,n-p-1}$.

Section 3

Partial F-tests

Motivating Example: CDI

- The dataset ***CDI*** gives various characteristics of 440 of the most populous counties in the USA.
- We'll try to predict y = unemployment percent from:
  - x_1 = percent of those over 25 with a BS degree;
  - x_2 = per capital income;
  - x_3 = percent of population over 65.
- The full model is

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i$$

CDI Example

```
cdi <- read.csv("data/CDI.csv")
summary(lm(result<-lm(unemployment~bachelor + capita_income + X65plus, data=cdi)))
```

```
##
## Call:
## lm(formula = result <- lm(unemployment ~ bachelor + capita_income +
##     X65plus, data = cdi))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0438 -1.1638 -0.2989  0.6977 12.8957
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    9.214e+00  5.275e-01  17.469   <2e-16 ***
## bachelor      -1.802e-01  1.955e-02  -9.219   <2e-16 ***
## capita_income  5.040e-05  3.468e-05   1.453    0.147
## X65plus        2.021e-02  2.694e-02   0.750    0.453
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.964 on 436 degrees of freedom
## Multiple R-squared:  0.2991, Adjusted R-squared:  0.2943
## F-statistic: 62.03 on 3 and 436 DF,  p-value: < 2.2e-16
```

CDI Example

- x_1 (BSDegree) is significant predictor of unemployment.
- x_2 (Income) is not significant when x_1 and x_3 are in the model.
- x_3 (Percentage of population over 65) is not significant when x_1 and x_2 are in the model.
- We do not know whether both x_2 and x_3 should be excluded after including x_1 in the model.

CDI Example

- In this situation, it is not interesting to conduct the overall F-test because
- We can still use F tests.
  - $H_0 : \beta_2 = \beta_3 = 0$.
  - H_1 : at least one of them is not zero.
- The reduced model (corresponding to H_0) is

$$y_i = \beta_0 + \beta_1 x_{i1} + \epsilon$$

CDI Example

```
lm.full<-lm(result<-lm(unemployment~bachelor + capita_income + X65plus, data=cdi))
lm.reduced<-lm(unemployment~bachelor, data=cdi)
anova(lm.reduced, lm.full)
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ bachelor
## Model 2: unemployment ~ bachelor + capita_income + X65plus
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    438 1697.5
## 2    436 1681.8  2     15.71 2.0365 0.1317
```

CDI Example

- The F-statistic is

$$F = \frac{(1697.5 - 1681.8)/(438 - 436)}{1681.8/436} = 2.0365$$

- The result suggest x_2 and x_3 are not needed in the model.

Section 4

Extra Sums of Squares

Extra Sums of Squares

- In multiple regression, we are often interested in how much one set of variables increases SSR (sum of squares for regression), given another set of variables already is in the model. The increment in SSR by adding a set of predictors (set 2) to an existing set of predictors (set 1) is

$$SSR(\text{set 2}|\text{set 1}) = SSR(\text{set 1 and set 2}) - SSR(\text{set 1})$$

or equivalently,

$$SSR(\text{set 2}|\text{set 1}) = SSE(\text{set 1}) - SSE(\text{set 1 and set 2})$$

Extra Sums of Squares: CDI Example

```
anova(lm.reduced)
```

```
## Analysis of Variance Table
##
## Response: unemployment
##            Df  Sum Sq Mean Sq F value    Pr(>F)
## bachelor    1  702.05  702.05  181.15 < 2.2e-16 ***
## Residuals 438 1697.47    3.88
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(lm.full)
```

```
## Analysis of Variance Table
##
## Response: unemployment
##                Df  Sum Sq Mean Sq  F value  Pr(>F)
## bachelor        1  702.05  702.05 182.0089 < 2e-16 ***
## capita_income   1   13.54   13.54   3.5099 0.06167 .
## X65plus         1    2.17    2.17   0.5631 0.45343
## Residuals     436 1681.76    3.86
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Extra Sums of Squares: CDI Example

```
anova(lm.reduced, lm.full)
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ bachelor
## Model 2: unemployment ~ bachelor + capita_income + X65pl
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    438 1697.5
## 2    436 1681.8  2     15.71 2.0365 0.1317
```

- In the CDI example, the extra sum of squares due to adding x_2 and x_3 to a model that already contains x_1 is

$$SSR(x_2, x_3|x_1) = SSE(x_1) - SSE(x_1, x_2, x_3) = 1697.5 - 1681.8 = 15.7$$

Partial F-test Using Extra Sums of Squares

- The partial F-test for testing $H_0 : \beta_2 = \beta_3 = 0$ vs. H_a : at least one of them is not zero can be conducted using extra sums of squares:

$$F = \frac{SSR(x_2, x_3|x_1)/2}{MSE(\text{full model})} = \frac{15.7/2}{1681.8/436} = 2.0365$$

- This is the same F-statistic as obtained before.

Decomposition of SSR

- Recall from SLR, we have

$$SST = SSR + SSE$$

where

$$SSR = \beta_1^2 \sum_{i=1}^{n} (x_i - \bar{x})^2$$

- In MLR with x_1 and x_2,

$$SST = SSR(\beta_1, \beta_2) + SSE(\beta_1, \beta_2)$$

and

$$SST = SSR(\beta_1) + SSR(\beta_2|\beta_1) + SSE(\beta_1, \beta_2),$$

or equivalently

$$SST = SSR(\beta_2) + SSR(\beta_1|\beta_2) + SSE(\beta_1, \beta_2)$$

Decomposition of SSR

- The order of predictors is aribitrary. For example, in a model with three predictors x_1, x_2, x_3, we have

$$\begin{aligned} SST &= SSR(x_1) + SSR(x_2|x_1) + SSR(x_3|x_1, x_2) + SSE(x_1, x_2, x_3) \\ &= SSR(x_2) + SSR(x_1|x_2) + SSR(x_3|x_1, x_2) + SSE(x_1, x_2, x_3) \\ &= SSR(x_3) + SSR(x_1|x_3) + SSR(x_2|x_1, x_3) + SSE(x_1, x_2, x_3) \\ &\cdots\cdots \end{aligned}$$

Example CDI

```
model12 <- lm(result<-lm(unemployment~bachelor + capita_income, data=cdi))
model21 <- lm(result<-lm(unemployment~capita_income + bachelor, data=cdi))
```

Example CDI: Model 12

```
anova(lm(result<-lm(unemployment~bachelor, data=cdi)))
```

```
## Analysis of Variance Table
##
## Response: unemployment
##            Df  Sum Sq Mean Sq F value    Pr(>F)
## bachelor    1  702.05  702.05  181.15 < 2.2e-16 ***
## Residuals 438 1697.47    3.88
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model12)
```

```
## Analysis of Variance Table
##
## Response: unemployment
##                Df  Sum Sq Mean Sq  F value  Pr(>F)
## bachelor        1  702.05  702.05 182.1910 < 2e-16 ***
## capita_income   1   13.54   13.54   3.5134 0.06154 .
## Residuals     437 1683.93    3.85
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example CDI: Model 12

- $SSR(\beta_1) = 702.05$
- $SSR(\beta_2|\beta_1) = 13.54$
- $SSE(\beta_1, \beta_2) = 1683.93$

Example CDI: Model 21

```
anova(model21)
```

```
## Analysis of Variance Table
##
## Response: unemployment
##                 Df  Sum Sq Mean Sq F value    Pr(>F)
## capita_income    1  249.02  249.02  64.622 8.567e-15 ***
## bachelor         1  466.58  466.58 121.082 < 2.2e-16 ***
## Residuals      437 1683.93    3.85
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example CDI: Model 21

- $SSR(\beta_2) = 249.02$
- $SSR(\beta_1|\beta_2) = 466.58$
- $SSE(\beta_1, \beta_2) = 1683.93$

Example CDI: Model 12 vs Model 21

- Note that $SSR(\beta_1, \beta_2) = SSR(\beta_1) + SSR(\beta_2|\beta_1) = SSR(\beta_2) + SSR(\beta_1|\beta_2)$
- The order of predictors entered in R depends on what hypothesis you are carrying out. We usually arrange the model so that the last coefficient are the ones to be tested.

Section 5

ANOVA vs t-test

Model 12 vs Model 12

```
summary(model12)
```

```
##
## Call:
## lm(formula = result <- lm(unemployment ~ bachelor + capita_income,
##     data = cdi))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0534 -1.1657 -0.3001  0.6941 12.8136
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    9.430e+00  4.420e-01  21.334   <2e-16 ***
## bachelor      -1.874e-01  1.703e-02 -11.004   <2e-16 ***
## capita_income  6.020e-05  3.212e-05   1.874   0.0615 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.963 on 437 degrees of freedom
## Multiple R-squared:  0.2982, Adjusted R-squared:  0.295
## F-statistic: 92.85 on 2 and 437 DF,  p-value: < 2.2e-16
```

Model 12 vs Model 12

```
summary(model21)
```

```
##
## Call:
## lm(formula = result <- lm(unemployment ~ capita_income + bachelor,
##     data = cdi))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0534 -1.1657 -0.3001  0.6941 12.8136
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    9.430e+00  4.420e-01  21.334   <2e-16 ***
## capita_income  6.020e-05  3.212e-05   1.874   0.0615 .
## bachelor      -1.874e-01  1.703e-02 -11.004   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.963 on 437 degrees of freedom
## Multiple R-squared:  0.2982, Adjusted R-squared:  0.295
## F-statistic: 92.85 on 2 and 437 DF,  p-value: < 2.2e-16
```

Model 12 vs Model 12

```
anova(model12)
```

```
## Analysis of Variance Table
##
## Response: unemployment
##                Df  Sum Sq Mean Sq  F value  Pr(>F)
## bachelor        1  702.05  702.05 182.1910 < 2e-16 ***
## capita_income   1   13.54   13.54   3.5134 0.06154 .
## Residuals     437 1683.93    3.85
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model21)
```

```
## Analysis of Variance Table
##
## Response: unemployment
##                Df  Sum Sq Mean Sq F value    Pr(>F)
## capita_income   1  249.02  249.02  64.622 8.567e-15 ***
## bachelor        1  466.58  466.58 121.082 < 2.2e-16 ***
## Residuals     437 1683.93    3.85
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANOVA (Partial F test) vs t-test

```
model1 <- lm(result<-lm(unemployment~bachelor, data=cdi))
model2 <- lm(result<-lm(unemployment~capita_income , data=cdi))
```

Partial F test: Model 12 vs Model 1

```
anova(model1, model12)
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ bachelor
## Model 2: unemployment ~ bachelor + capita_income
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1    438 1697.5
## 2    437 1683.9  1    13.539 3.5134 0.06154 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Partial F test: Model 21 vs Model 1

```
anova(model1, model21)
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ bachelor
## Model 2: unemployment ~ capita_income + bachelor
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1    438 1697.5
## 2    437 1683.9  1    13.539 3.5134 0.06154 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Partial F test: Model 21 vs Model 2

```
anova(model2, model21)
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ capita_income
## Model 2: unemployment ~ capita_income + bachelor
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    438 2150.5
## 2    437 1683.9  1    466.58 121.08 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model 12 vs Model 2

```
anova(model2, model12)
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ capita_income
## Model 2: unemployment ~ bachelor + capita_income
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    438 2150.5
## 2    437 1683.9  1    466.58 121.08 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

t-test

```
summary(lm(result<-lm(unemployment~bachelor + capita_income, data=cdi)))
```

```
##
## Call:
## lm(formula = result <- lm(unemployment ~ bachelor + capita_income,
##     data = cdi))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0534 -1.1657 -0.3001  0.6941 12.8136
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    9.430e+00  4.420e-01  21.334   <2e-16 ***
## bachelor      -1.874e-01  1.703e-02 -11.004   <2e-16 ***
## capita_income  6.020e-05  3.212e-05   1.874   0.0615 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.963 on 437 degrees of freedom
## Multiple R-squared:  0.2982, Adjusted R-squared:  0.295
## F-statistic: 92.85 on 2 and 437 DF,  p-value: < 2.2e-16
```

Conclusion

- The partial F-test and the t-test give the same conclusion regarding the significance of individual predictors in the MLR model.
- The predictor to be test should be the last one(s) entered in the model for the partial F-test.
- Both tests are based on the same extra sums of squares and assesses the same null hypothesis - after adjusting for other predictors in the model, is the predictor still significant.